

Do you need AES-256 level safety?

AES standard comes in three bit sizes for security: 128, 192 and 256. While a number of developers opt for AES 256-bit for better safety, 128-bit version itself is strong enough for general-purpose IoT systems. AES-128 implements a 128-bit key to encrypt the data using 10 rounds of ciphering. Theoretically, the 128-bit key can be recovered by performing 2^{128} trials of brute force attack—which is an almost unachievable feat. Similarly, 192-bit encryption uses a 192-bit key with 12 rounds of ciphering, and AES-256 uses 256-bit key with 14 rounds of ciphering. So, AES-256 is theoretically recoverable by 2^{256} trials of a brute attack, which is an impossible time-frame. AES-192 and AES-256 have been designed for federal- and military-level system security.

efficiencies, drive operational performance to enable innovation, and employ advanced data analytics to transform businesses. With its layered approach towards multiple security features, and design patterns, Azure IoT Suite helps deploy an infrastructure that can be trusted by any business."

What's the future

In the coming two to three years, even newer security upgrades will

be available and the systems will have to be updated. Code signing abilities with higher levels of encryption may be a step forward to secure the devices.

As regards security practices for the IoT, Dr Bhatnagar said, "A security operations centre needs to be established to monitor threats in a proactive manner while detecting and tracking signatures of divergent activities. On the tactics side, sanitise your operational environment by deploying

patches, keeping segmented offline back-ups, conducting audits and deriving visibility into your core and peripheral networks. If a malware breaches your first layer of defence, you should be able to know where it is and what exactly it is doing. Most importantly, ensure that everyone in your organisation understands the collective responsibility they bear for cybersecurity."

Experts feel that endpoint security monitoring will see the maximum advancements in the next ten years. Data analytics, machine learning and artificial intelligence will play a big role in prediction of attacks and suspicious activities. The current movement towards a common IoT standard will influence the future security trends. We will see better encryption capabilities and more evolved authentication and authorisation processes. **EPY**



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